GIDFINE LOAN PLATFORM

PHASE 4A CORRECTION PASS & REMEDIATION REPORT

Audit Remediation & Idempotency Hardening | Branch: development | HEAD: 931411f

Date: October 21, 2026	**Target Branch:** development (HEAD @ 931411f)
Execution Mode: Limited Implementation (Remediation)	**main Branch:** Untouched (at f7ed8d1)
Audit Findings Resolved: F-01, F-02, F-04, Concurrency	**Validation Status:** 100% Passing (44 Node + 6 Py + Vite)

1. Executive Summary & Preflight Status

The Phase 4A Correction Pass was executed in strict accordance with the limited implementation boundaries. All verified audit findings (F-01 documentation inconsistency, F-02 optional idempotency key guard, and F-04 payload-mismatch test) have been resolved without introducing Phase 4B business mutations or altering the core database schema. The branch is confirmed as development at HEAD commit 931411f. Zero commits were created, zero pushes were executed, and main remains untouched at f7ed8d1.

2. Addressed Audit Findings & Remediation Details

Finding ID	Remediation Summary	Code / Doc Location
F-01 Documentation	Rebuilt Phase 4A Implementation PDF. Explicitly listed idempotencyKey: optional string in the ledgerEntries field table, ensuring exact consistency with schema.ts:215.	`migration/GIDFINE_PHASE4A_LEDGER_SCHEMA_REPORT.pdf`
F-02 Optional Key Guard	Implemented isValidIdempotencyKey helper. Established contract: records without an idempotency key are stored as undefined and never matched against missing/empty keys on query.	`frontend/convex/lib/ledgerTypes.ts:193`
F-04 Payload Mismatch	Implemented evaluateIdempotencyPayloadMatch helper and unit tests. Reusing an existing idempotency key with differing amounts or application IDs is strictly rejected.	`frontend/convex/lib/ledgerTypes.ts:205` `frontend/tests/ledgerSchema.test.ts:257`
Concurrency OCC Claims	Clarified three-tier model: a) Schema indexes lack unique constraints; b) Convex OCC serializes concurrent writes via read-set invalidation; c) Concurrency safety depends on Phase 4B mutation lookups.	PDF Section 3 & Test Documentation

3. Validation & Test Execution Results

Target Test Suite	Command Line	Exit Code	Status
TypeScript Static Analysis	`npx tsc --noEmit`	0	**PASSED (0 errors)**
Node Test Runner (All Suites)	`node --experimental-strip-types --test tests/*.test.ts`	0	**PASSED (44 / 44 tests)**
Phase 4A Focused Suite	`node --test tests/ledgerSchema.test.ts`	0	**PASSED (14 / 14 tests)**
Python Backend Tests	`pytest tests/ -q`	0	**PASSED (6 / 6 tests)**
Production Vite Bundle	`npx vite build`	0	**PASSED (1,845 modules)**

4. Remaining Limitations & Phase 4B Dependencies

• **Mutation Runtime Verification:** Idempotency payload matching and Maker-Checker segregation are verified via pure TypeScript helper functions and unit tests. They will be integrated and verified within live Convex mutation transactions in Phase 4B.
• **Double-Entry Accounting:** The schema remains an append-only transaction sub-ledger tied to loan applications; it does not model balanced general ledger debit/credit accounts.
• **Maker-Checker Enforcement:** No approval mutations were added. Authenticated server-side enforcement (checkerUser._id !== makerId) remains a Phase 4B requirement.

5. Strict Scope & Boundaries Confirmation

• **Branch Safety:** All work remained strictly on development. main was untouched at f7ed8d1.

• **Zero Commits:** No Git commits or branch pointer changes were created in this correction pass.

• **Zero Remote Operations:** Zero pushes executed; Convex Cloud remains undeployed.

• **Zero Business Logic:** No disbursement, approval, ledger posting, or repayment mutations were added.

• **Final Status:** STOPPED after Phase 4A correction pass. Ready for review before Phase 4B.